

Dr. Elias J. Thorne

Ballerup, Denmark

in <https://linkedin.com/in/eliasthorne>

G <https://github.com/eliasthorne-ai>

+45 12 34 56 78

elias.thorne@email.com

Summary

Accomplished AI Engineer with over six years of experience designing, developing, and deploying frontier AI applications across healthcare, enterprise software, and cloud-native environments. Proven expertise in translating cutting-edge research into production-grade solutions, with deep hands-on experience in LLMs, vector databases, semantic search, and agentic workflow architectures. Adept at collaborating with data scientists, domain experts, and product teams to reimagine business processes through structured AI frameworks and accelerator methodologies. Highly proficient in modern development practices, including CI/CD pipelines on GitHub, cloud infrastructure (AWS, Azure, Databricks, Snowflake), and AI governance standards. Passionate about leveraging AI to solve complex clinical and operational challenges, driving measurable impact while maintaining rigorous quality, security, and ethical standards. Eager to bring rapid prototyping capabilities, continuous learning, and cross-functional leadership to Novo Nordisk's Forward Deployed AI Engineering team, contributing directly to the mission of #BringDataToLife and advancing patient care globally.

Work Experience

- MedTech AI Solutions** Ballerup, Denmark
Senior AI Engineer 2021-06 - Now
 - Led end-to-end development of LLM-powered clinical information retrieval systems, utilizing vector databases and semantic search to improve data accessibility for medical professionals.
 - Collaborated directly with data scientists and clinical domain experts to translate research prototypes into scalable, production-ready AI applications.
 - Designed and implemented agentic evaluation frameworks using OpenTelemetry and MCP standards, enabling continuous feedback loops and performance benchmarking.
 - Integrated AI solutions into existing engineering teams, establishing best practices for model versioning, observability, and responsible AI deployment.
 - Reduced model inference latency by 35% through architectural optimizations and cloud-native scaling strategies on AWS and Azure.
 - Mentored junior engineers in modern IDE integrations (Cursor, VS Code), prompting best practices, and agile AI development methodologies.
- CloudScale Enterprise** Ballerup, Denmark
AI Platform Engineer 2019-03 - 2021-05
 - Co-built and maintained enterprise AI platform components processing billions of tokens monthly, focusing on scalability and reliability.
 - Developed reusable AI infrastructure and evaluation benchmarks for generative models, ensuring alignment with business-critical accuracy requirements.
 - Implemented CI/CD pipelines using GitHub Actions to automate model training, testing, and deployment, accelerating release cycles by 40%.
 - Partnered with Product AI Engineers to generalize bespoke prototypes into standardized, company-wide services.

- Supported the AI Accelerator framework by mapping business use cases to technical specifications and monitoring post-deployment impact.

- **DataDriven Innovations** Ballerup, Denmark
AI & Data Engineer 2017-09 - 2019-02
 - Designed and deployed machine learning pipelines for predictive analytics and natural language processing tasks.
 - Integrated semantic search capabilities into enterprise knowledge bases, enhancing information retrieval for cross-functional teams.
 - Utilized rapid prototyping techniques and modern coding assistants to validate AI hypotheses before full-scale implementation.
 - Contributed to full-stack AI product delivery, from ideation and data engineering to model deployment and stakeholder communication.

Projects

- **Enterprise RAG & Semantic Search Platform** 2022-01 - 2023-06
Lead AI Architect
 - Architected a hybrid Retrieval-Augmented Generation (RAG) system combining vector stores, knowledge graphs, and LLMs for precise semantic query resolution.
 - Integrated OpenTelemetry for end-to-end observability of LLM inference pipelines, tracking latency, token usage, and error rates.
 - Delivered a production-ready prototype that reduced enterprise information retrieval time by 65% and improved answer accuracy by 28%.
 - Established evaluation benchmarks and feedback mechanisms to continuously refine retrieval strategies and model outputs.
- **Agentic Workflow Automation for Clinical Data Processing** 2023-07 - 2024-04
AI Framework Designer
 - Developed an autonomous agent orchestration layer utilizing A2A protocols and MCP standards to automate multi-step clinical data validation workflows.
 - Leveraged VS Code Dev Containers and Cursor for rapid iteration, testing, and deployment of agent skills and tooling.
 - Piloted the system across three cross-functional teams, improving data pipeline efficiency by 30% and reducing manual review overhead.
 - Documented architecture decisions and integration patterns to facilitate future scaling and replication across business units.

Education

- **Imperial College London** Ballerup, Denmark
PhD in Computer Science (Artificial Intelligence & Machine Learning) 2015-09 - 2019-06
- **Technical University of Denmark (DTU)** Ballerup, Denmark
MSc in Data Science & Artificial Intelligence 2013-08 - 2015-06

Courses & Certifications

- **Advanced LLM Architectures & Prompt Engineering** Stanford University
2023-11
- **AI Governance, Ethics & Responsible AI Practices** MIT Professional Education
2022-08
- **Cloud-Native AI Platforms on AWS & Azure** AWS Training & Certification
2021-05
- **Advanced CI/CD for Machine Learning Workflows** Coursera / DeepLearning.AI
2020-03

Technical Skills

Python, PyTorch, TensorFlow, JAX, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (Pinecone, Milvus, Weaviate, Chroma), Semantic Search, MCP (Model Context Protocol), A2A (Agent-to-Agent), Agent Skills & Orchestration, OpenTelemetry, Docker & Kubernetes, CI/CD (GitHub Actions, Jenkins, GitLab CI), AWS (SageMaker, Bedrock, ECS, Lambda), Azure (AI Studio, ML Ops), Databricks, Snowflake, Git & Version Control, Agile & Scrum Methodologies, Prompt Engineering & Chain-of-Thought, Model Evaluation & Benchmarking, Responsible AI & Governance, Technical Documentation & Communication, Cross-Functional Collaboration, Rapid Prototyping & Innovation

Languages

English: Native

Danish: Professional Working Proficiency

German: Conversational

Patents

- **System and Method for Dynamic Semantic Query Resolution** US2023123456A1
in Healthcare Data
Pending 2023-04

Publications

- **Optimizing RAG Pipelines for Clinical NLP Tasks: A Comparative Study** 2023-11
- **Agentic Frameworks in Enterprise AI Deployment: Architecture, Evaluation, and Governance** 2024-02